Product Requirements Document — Elder Link

Field	Value
Product	Elder Link / HK Elderly AI Check-In Call
Owner	Ali — Software Owner; product and business inputs jointly owned by the named team
Status	Draft
Last Updated	8 October 2026
Vision	Help Hong Kong elders living alone stay connected and be checked regularly through safe Cantonese calls that route concerns to people who can help.
Primary success metric and target	**Prototype:** Safe-response, worrying-response, and no-answer journeys each pass twice consecutively; required human escalation and sensor-triggered calling begin within one minute.

> **Document convention:** Unconfirmed statements are marked **Estimate**, **Assumption**, **Recommendation**, or **Open Question**. "Prototype" means the six-day demonstration using sample elder records and consenting team test numbers. "Pilot" means a future controlled trial with real elders.

1. EXECUTIVE SUMMARY

Product vision

Elder Link is a phone-based support workflow for Hong Kong residents aged 65+ who live alone, especially hidden elders. It places automated Cantonese calls every two to three days to check whether an enrolled elder is safe. A safe response is logged and the elder hears one suitable centre activity. A request for help, an unsafe statement, an unclear or confused answer, or repeated no-answer triggers an automated voice call through a defined human contact chain. The same workflow can verify an existing sensor alert, helping staff distinguish a confirmed safe response from a case requiring follow-up.

The elder needs no app, smartphone, wearable, camera, or always-on microphone: the service calls ordinary mobile phones and landlines. It does not diagnose, give medical advice, replace 999 or an emergency alarm, continuously detect falls, or make emergency decisions. A human decides every emergency action.

The key differentiator is the combination of: (1) automated Cantonese check-in conversations, (2) verification of sensor alerts, (3) human escalation, and (4) community-service connection. This is, **to the team's current knowledge and subject to validation**, an uncommon combination in Hong Kong; the PRD does not claim definitively that it is the first.

Strategic alignment

Objective	Product contribution	Evidence to collect
No enrolled elder goes unchecked for more than about three days	Scheduled calls every 2–3 days plus retry and escalation	Schedule coverage and overdue-check reports
Reduce wasted frontline checks without hiding risk	AI verifies routine status and sensor alerts; humans handle ambiguity and risk	Safe-closure rate, escalations, staff workload feedback, false-safe review
Connect elders to community	Safe calls mention one appropriate centre activity and pass stated interest to staff	Activity-offer, interest, and completed follow-up events
Reduce isolation and solitary-death risk	Repeated familiar contact, faster follow-up, and community connection	**Assumption:** contribution is plausible but must not be presented as causal until evaluated
Validate a public-service model	Controlled pilot measures safety, adoption, workload, connection, and cost	Government/stakeholder pilot report

Market opportunity and timing

The supplied evidence identifies 188,569 Hong Kong residents aged 65+ living alone in 2021, sensor pilots covering roughly 950 households, around 215 elderly centres, and no official solitary-death count. These figures establish a substantial target group and existing care infrastructure, but do not prove demand, cost savings, or mortality impact.

The first intended payer is the Hong Kong government; the elder never pays. The first real pilot is planned for 20–30 eligible elders. Pricing, charging unit, budget owner, procurement route, and deployment price remain **Open Questions**. The team will validate operating costs and government buyer needs before estimating a future range.

Resource requirements

Resource	Availability / responsibility
Ali	Software owner; architecture, implementation, repository, deployment, technical readiness
Eden and Jovian	Sensor simulation/integration, telephony and scenario testing
Abebi and Jayson	User research, business and social-value content, activity content, demo story
Capacity	2–4 hours/person/day for six days; **Estimate:** 60–120 person-hours total
Current assets	Problem research, early AI development, possible ESP32 simulation, limited personal access to one elderly centre
Tools	Shared GitHub repository, WhatsApp, small paid services approved case by case

Prototype constraint	Sample data and consenting team test numbers only; no real elder records

Prototype, pilot, and exclusions

Six-day prototype: real calls to team test numbers; safe, worrying, no-answer, and sensor-triggered flows; one retry; voice escalation chain; short notes without audio; responsive bilingual dashboard; separate logins; test evidence; recorded backup demonstration.

Future pilot: 20–30 real elders only after target-user research, consent, provider validation, legal/privacy/telecommunications/safeguarding review, operating ownership, staff training, and incident procedures are approved.

Out of V1: non-Cantonese languages; diagnosis or medical advice; replacement of 999/emergency alarms; continuous fall detection; payments, financial data or Hong Kong ID collection; automated emergency decisions; use for moderate/severe dementia, residential-care residents, severe hearing loss, or non-Cantonese speakers; retained call audio.

2. PROBLEM STATEMENT AND OPPORTUNITY

Problem definition

Hong Kong elders living alone may lack frequent, reliable human check-ins. Frontline staff and volunteers have limited capacity, while sensor alerts can create unnecessary follow-up when an elder is safe. Existing communication channels may also fail to connect hidden elders with suitable centre activities.

Elder Link addresses three linked gaps:

1. **Coverage:** every enrolled elder should receive a check at least about every three days.
2. **Triage:** a short Cantonese call distinguishes a safe response from concern, ambiguity, or no answer, while reserving emergency judgment for a human.
3. **Connection:** a safe interaction can introduce one appropriate community activity and pass interest to staff.

Quantified evidence and limitations

- 188,569 people aged 65+ lived alone in Hong Kong in 2021.
- Sensor pilots cover roughly 950 households.
- Around 215 elderly centres are cited in the source materials.
- No official solitary-death count is available in the supplied research.
- Existing evidence does not yet quantify willingness to answer AI calls, actual staff-hours saved, operating cost, or safety outcomes.

User segments

- **Primary:** Cantonese-speaking Hong Kong residents aged 65+ living alone, particularly hidden elders who remain outside regular services.
- **Secondary:** Care Team volunteers, elderly-centre social workers, and alert-service operators.
- **Affected stakeholders:** designated contacts, family members, neighbours, care

organisations, housing providers, government funders, privacy/legal/safeguarding owners, and telephony providers.

Competitive gap

The proposed advantage is not a single technology. It is the integrated workflow of regular Cantonese calls, sensor-alert verification, human escalation, and community-service referral. The "first" or "unique" positioning remains subject to validation and must not be used as an established fact.

Business opportunity

The first buyer focus is the Hong Kong government. A real pilot should prove safety and staff-workload value, grow elder enrolment, and strengthen community connection. Commercial model and price are intentionally open until the team measures actual telephony, hosting, staffing, support, compliance, and procurement requirements.

Success criteria

Prototype: all required scenarios pass twice consecutively; one-minute initiation targets are met; no real elder data or retained audio; dashboard workflow and evidence bundle complete.

Pre-pilot research: at least five elders and two staff; at least 70% of elders willing to try after in-person introduction (**Recommendation:** at least four of five in the minimum sample); most staff expect lower unnecessary workload without hiding risk.

Pilot: measure check coverage, timing breaches, false-safe/missed events, staff workload, acceptance/opt-out, activity connection, provider performance, and actual cost. Exact production thresholds require stakeholder approval.

Stakeholder alignment points

Before real use, stakeholders must agree on the eligible cohort, meaning of "checked" and "safe," operator roster and service hours, consent evidence, permissions, approved scripts, 999 wording, opt-out and deletion, false-safe review, activity ownership, provider and caller-number rules, incident/stop criteria, liability, procurement, and funding.

3. USER REQUIREMENTS AND STORIES

Personas

Persona	Goals	Current pain points	Success criteria
Elder living alone	Dignified Cantonese contact at a preferred time; easy reschedule/opt-out; trusted identity	Scam fear, unfamiliar callers, isolation, no desire for a new device	Understands who is calling, completes short check-in, knows service never asks for money/ID, can opt out
Care Team volunteer	Know which cases genuinely need attention	Repeated manual calls, incomplete context, alert fatigue	Clear urgent queue, current status, simple acknowledgement and

			handoff
Social worker	Safe triage and service referral context	Fragmented records, limited time, concern that automation hides risk	Auditable history, activity interest, human control over emergency action
Service operator	Reliable escalation and operating visibility	Unanswered contacts, inconsistent procedures, peak workload	Defined chain, timestamps, unresolved-case warning
Designated contact	Confidence that a concern reaches a person	Unclear responsibility and delayed information	Consented escalation route and clear notification
Government payer	Evidence of safety, adoption, workload value, privacy, and cost	Unproven outcomes and procurement risk	Controlled pilot with transparent limits and measurable results

Current journey

1. Check-ins depend on available staff or volunteers and may be irregular.
2. Sensor alerts require a person to verify whether help is needed.
3. No-answer follow-up may be inconsistent or difficult to audit.
4. Welfare checking and community-activity promotion are separated.
5. **Assumption:** records and ownership are fragmented; staff research must validate this.

Proposed future journey

1. Staff enrol an eligible elder after the elder, one designated contact, and the referring organisation agree.
2. The service is introduced in person, including the fixed number, AI identity, privacy boundaries, rescheduling, and opt-out.
3. Elder Link calls at the preferred time every 2–3 days or within one minute of a sensor alert.
4. A safe answer creates a short note and offers one suitable activity. A safe sensor case also informs the designated human/contact.
5. A worrying or unclear answer escalates within one minute. Immediate danger prompts advice to call 999 and immediate human escalation; Elder Link does not claim to contact 999.
6. No answer leads to one retry after 5–10 minutes; a second no-answer escalates within one minute.
7. The voice escalation chain calls first contact, second contact, then operator.
8. Authorised staff acknowledge, follow up, and record a short note. Humans decide emergency action.
9. Notes remain while the elder is enrolled and are deleted within 30 days after exit.

Epics and user stories

Epic A — Consent and enrolment

- As care staff, I want to register a minimal profile and three-party consent so that only agreed participants enter the service.
- As an elder, I want to choose my call time, reschedule, and opt out so that the service respects my autonomy.

Epic B — Scheduled Cantonese check-in

- As an elder, I want a short Cantonese call on my ordinary phone so that I can confirm safety without learning a new device.
- As care staff, I want every active elder scheduled every 2–3 days so that missed checks become visible.

Epic C — Safe, worrying, and no-answer handling

- As an elder who feels unsafe or asks for help, I want a human alerted quickly.
- As an operator, I want repeated no-answer escalated through a defined chain.
- As staff, I want safe outcomes logged without full audio or transcripts.

Epic D — Sensor verification

- As a responder, I want a sensor alert to start a check call within one minute.
- As a designated contact, I want notification even when a sensor-triggered call confirms safety.

Epic E — Community connection

- As an elder who is safe, I want to hear one suitable activity without pressure.
- As a social worker, I want stated interest recorded for follow-up.

Epic F — Shared dashboard

- As an authorised volunteer, social worker, operator, or named tester, I want my own login and one consistent view.
- As staff, I want to acknowledge an alert, add a short note, and mark follow-up complete.

4. FUNCTIONAL REQUIREMENTS

MoSCoW prioritisation

ID	Requirement	Priority	Impact	Effort
FR-01	Minimal sample elder profile, eligibility, and	Must	High	Medium

	consent gate			
FR-02	Schedule every 2–3 days and overdue visibility	Must	High	Medium
FR-03	Real Cantonese call to ordinary phone, AI disclosure, safe script	Must	High	High
FR-04	Safe/worrying/unclear/immediate-danger/no-answer classification	Must	High	High
FR-05	Safe note, one activity offer, interest capture	Must	Medium	Medium
FR-06	Worrying/immediate-danger escalation within one minute	Must	High	Medium
FR-07	No-answer retry after 5–10 minutes, then escalation	Must	High	Medium
FR-08	Sensor event starts call within one minute; safe closure and notification	Must	High	Medium
FR-09	First contact → second contact → operator	Must	High	Medium
FR-10	Separate logins and authorised shared dashboard	Must	High	Medium
FR-11	Acknowledge, complete, note, history, and audit	Must	High	Medium
FR-12	Traditional Chinese/Cantonese wording + English; phone/desktop; accessible labels	Must	High	Medium
FR-13	No audio; exit deletion within 30	Must	High	Low

	days			
FR-14	Failure retry and recorded backup demonstration	Must	Medium	Low
FR-15	Elder-record and alert/case adapter interfaces	Should	Medium	Medium
FR-16	Reschedule and opt-out during call	Should	High	Medium
FR-17	Larger activity catalogue and matching	Could	Medium	Medium
FR-18	Reminder and post-activity calls	Could/Future	Medium	Medium
FR-19	Live partner integration	Won’t unless safe and non-disruptive	Medium	High
FR-20	Additional languages, diagnosis, payments, continuous sensing	Won’t in V1	Out of scope	High

Must-Have acceptance criteria

FR-01 — Enrolment, eligibility, and consent

- **Given** any required field or consent from the elder, designated contact, or organisation is missing, **when** activation is attempted, **then** activation is blocked and the missing item is named.
- **Given** all minimum sample fields and three agreements are present, **when** staff activate the profile, **then** a next-check window is created.
- **Given** the profile is outside the V1 cohort, **when** activation is attempted, **then** it is blocked with an eligibility explanation.

Minimum sample fields: reference number, preferred name, phone, preferred call time, two contacts, centre, language, and consent status. Prototype records are fictitious and phone numbers belong only to consenting team testers.

FR-02 — Scheduling and coverage

- **Given** an active profile, **when** the configured two- or three-day interval arrives, **then** a call job is created by the due time plus a **Recommended** 60-second scheduler tolerance.
- **Given** no completed check by the end of the window, **when** staff open the dashboard, **then** an overdue text label and icon appear.
- **Given** a rescheduled time, **when** it is saved, **then** the next job and audit history update.

FR-03 — Real Cantonese calling

- **Given** a consenting team mobile or landline, **when** a check starts, **then** the phone rings and the caller hears understandable Cantonese service and AI disclosure.
- **Given** the approved script, **when** reviewed, **then** it requests no money, bank information, payment, HKID, diagnosis, or medical advice.
- A fixed caller number is required for real pilot use; vendor support must be proven during the Day 1 spike.

FR-04 — Outcome classification

- **Given** a clearly safe test phrase, **then** outcome is `SAFE` and no AI emergency decision is made.
- **Given** a request for help, an unsafe statement, or an unclear/confused phrase, **then** the case enters escalation.
- **Given** immediate danger, **then** the caller advises the elder to call 999 and human escalation starts without claiming that Elder Link called 999.
- **Given** no connected human response, **then** the result is `NO_ANSWER`, never safe.

FR-05 — Safe outcome and activity

- **Given** a safe response, **then** the latest status and time are saved in a short note and no audio is retained.
- Exactly one approved activity is mentioned.
- If the caller expresses interest, an activity-interest event becomes visible to staff.

FR-06 — Worrying escalation

- **Given** a worrying or unclear result at time T, **then** the first human voice call starts by T+60 seconds and timestamps are saved.
- The dashboard shows the escalation and current chain step.
- For immediate danger, 999 guidance and human escalation occur immediately according to the technically validated call behavior.

FR-07 — No-answer workflow

- After first no-answer, exactly one retry is scheduled 5–10 minutes later.
- If the retry connects, the resulting safe or worrying path runs without duplicate escalation.
- If the retry is also unanswered at T, the first human escalation call starts by T+60 seconds.

FR-08 — Sensor verification

- A valid simulated sensor event at T starts the elder call by T+60 seconds.
- A safe result closes the alert with a short safe note and initiates notification to the designated human/contact.
- Worrying or no-answer outcomes use the standard escalation chain and remain visible.

FR-09 — Contact chain

- If first contact answers, later contacts are not called and acknowledgement is logged.
- If first contact does not answer, second contact is called; if second does not answer, the operator is called.
- If all fail, the case remains visibly unresolved for manual handling.

FR-10 — Authentication and shared view

- Valid users sign in separately and see only the authorised shared workflow.
- Invalid or expired credentials reveal no elder data.
- Three simultaneous authorised users can use the prototype without losing updates.

FR-11 — Dashboard case work

- The list/detail displays elder reference, latest result/time, escalation status, and recent history, newest first.
- Acknowledgement, follow-up completion, and short-note actions save actor and timestamp.
- No audio or unsupported sensitive field is exposed.

FR-12 — Bilingual, responsive, accessible UI

- Core navigation, status, actions, and errors work in Traditional Chinese with natural Cantonese wording and in English.
- At representative phone and desktop widths, essential actions require no horizontal scrolling.
- Every colour state also has text and an icon; labels and keyboard focus are visible.

FR-13 — Retention and no audio

- Inspection after calls finds no audio file or recording URL.
- Within 30 days after exit, the elder profile and all call/follow-up notes are permanently deleted. Only non-reconstructive audit metadata (for example, deletion completion time and responsible system process) may remain, subject to legal/privacy approval.
- The remaining audit record must not recreate note content.

FR-14 — Demo failure handling

- A live failure triggers one retry and is recorded.
- A second failure triggers the dated backup recording; the presenter discloses that the live path failed and makes no live-success claim.

Definition of Done

A feature is Done only when it:

1. meets its acceptance criteria;
2. works in the integrated prototype;

3. is independently checked by someone other than its author;
4. has automated tests where practical and a completed manual checklist;
5. records timestamps, screenshots, reviewer, and known issues;
6. contains no real elder data or unapproved number;
7. retains no audio;
8. passes affected bilingual and accessibility checks;
9. demonstrates error/fallback behavior;
10. updates relevant API, schema, and run instructions;
11. receives one other technical member's review before merge; and
12. receives joint business and technical approval for demonstration.

5. TECHNICAL REQUIREMENTS

Recommended architecture and stack

Layer	Recommendation	Plain-language justification
Staff web UI	React + TypeScript responsive web app	One codebase can serve phone and computer browsers quickly.
API/workflow	Small Node.js + TypeScript service	One language across front and back simplifies a six-day build.
Database	Managed PostgreSQL-compatible store	Relationships and timestamped history are easier to validate and audit.
Jobs	Managed scheduler or lightweight durable queue	Needed for two-day scheduling, 5–10 minute retry, and one-minute escalation.
Calls/speech	Managed provider that proves Hong Kong outbound calling and Cantonese capability	Building telephony is outside the six-day scope.
Testing	Unit/API tests plus scripted real-call scenarios	Provides repeatable regression checks and real evidence.
Hosting	Managed test hosting	Reduces deployment and maintenance work during the sprint.

Open Question: no telephony vendor is confirmed. Day 1 must verify Hong Kong mobile and landline calling, account approval, fixed caller number, Cantonese quality, webhook behavior, disabled recording, data routing, cost controls, and applicable telecommunications requirements.

Components and data flow

- Enrolment/consent validator
- Scheduler and overdue monitor
- Call orchestrator and provider webhook handler

- Conservative outcome classifier with hard safety rules
- Escalation engine and contact chain
- Sensor and future partner adapters
- Activity service
- Authenticated case API and dashboard
- Retention/deletion worker
- Audit, analytics, and health monitoring without audio or transcript storage

Scheduled flow: due job → real call → safe/worrying/no-answer classification → note/activity or retry/escalation → dashboard/audit → next schedule.

Sensor flow: unique sensor event → call within one minute → safe closure plus designated-human notification, or standard escalation.

Integrations

Integration	Prototype	Future pilot
Calls/speech	One validated provider; team numbers only	Contracted provider with approved Cantonese, caller ID, landline coverage, no recording, and data routing
Elder records	Sample JSON/CSV or internal fake records behind adapter	Partner connector after access agreement and field mapping
Sensor/alert	Button, mock event, or isolated ESP32 simulation	Authenticated webhook/device integration with replay protection
Case management	Internal case table and sample outbound adapter	Bidirectional partner integration and ownership rules
Activities	Small approved sample dataset	Centre-owned catalogue with availability and suitability controls

One live partner connection may be added only if access unexpectedly becomes available and it does not threaten core acceptance.

Proposed APIs

Method / endpoint	Purpose
`POST /v1/auth/login`	Separate staff/tester login
`GET /v1/elders`	Authorised shared list with status/overdue filters
`POST /v1/elders`	Create sample or future enrolment record
`PATCH /v1/elders/{id}`	Update allowed fields, call time, or exit
`POST /v1/elders/{id}/activate`	Activate after consent and eligibility checks
`POST /v1/check-ins`	Start a scheduled or demo call job

`GET /v1/cases/{id}`	Retrieve current case, escalation, history, and notes
`POST /v1/sensor-alerts`	Accept unique simulated/future partner alert
`POST /v1/telephony/events`	Receive verified provider events idempotently
`POST /v1/cases/{id}/acknowledge`	Save staff acknowledgement
`POST /v1/cases/{id}/follow-up`	Complete follow-up and short note
`POST /v1/elders/{id}/opt-out`	End participation and set deletion due date
`GET /v1/analytics/summary`	Aggregate prototype/pilot metrics

All protected endpoints use HTTPS, authenticated identity, server-side role checks, request IDs, structured errors, version conflict checks, and audit events. Provider/sensor events use signatures or prototype shared secrets plus replay protection.

Recommendation: staff API 60 requests/minute/user; login 5 failed attempts/15 minutes; call creation 3/minute/elder. These are prototype safeguards, not validated production limits. Use bounded retry only for transient errors and idempotency keys to prevent duplicate calls/escalations.

Data requirements

Core entities: users; elders; contacts; consents; check-in schedules; call attempts; sensor alerts; cases; escalations; escalation attempts; activities; activity interests; notes; audit events; analytics events.

Rules:

- `elders`: minimal profile, status, enrolment/exit/deletion dates; no HKID, payment, or financial fields.
- `contacts`: first, second, and service-level operator with consent and priority.
- `call_attempts`: timestamps, status, outcome, and short note only; no audio, recording URL, or full transcript.
- `sensor_alerts`: unique external event ID, source, status, short safe note, human-informed time.
- `cases`: trigger, current outcome, escalation, acknowledgement, completion, and version.
- `audit_events`: append-only actor/action/time/request metadata that does not copy deleted note contents.
- `analytics_events`: pseudonymous references and categories, never phone numbers, audio, transcript, or note text.

Security and privacy

- TLS in transit and managed encryption at rest.
- Secrets outside source control; separate test and future pilot environments.
- Least privilege and API-side role enforcement.
- Mask phone numbers in list views and logs.
- Verify webhooks, protect login, sanitise input, scan dependencies, and audit record-changing actions.
- Three-party consent is checked at activation and before calls.
- No audio retention; notes remain only while enrolled and are deleted within 30 days after

exit.

- Future real records should be hosted in Hong Kong where practical and approved.
- Required before real use: legal, privacy, safeguarding, and Hong Kong telecommunications review. This PRD is not legal advice.

Performance and scalability

Prototype targets: up to three concurrent dashboard users; **Recommended** p95 dashboard response under two seconds with sample data; sensor-to-call and concern-to-escalation start within 60 seconds; retry after 5–10 minutes; schedule every 2–3 days.

Future pilot: initially 20–30 elders. Add organisation tenancy, duty rosters, production resilience, adapter versioning, language packs, and capacity planning only after pilot evidence.

6. USER EXPERIENCE REQUIREMENTS

Design principles

- Respectful, transparent, calm, non-clinical, and anti-scam by design.
- Use short natural Cantonese sentences, slower pacing, and one question at a time.
- Name the service and disclose AI at the start.
- Never request money, payment, bank details, or HKID.
- Repeat/rephrase an unclear answer once, then escalate rather than guess.
- For immediate danger, use approved wording: advise the elder to call 999 and state that a human contact is being alerted.
- Mention only one suitable activity during a safe call; do not pressure the elder.
- **Recommendation:** routine call under three minutes, subject to elder research.

Screens and navigation

Screen	Content/actions
Sign in	Language selection, credentials, accessible errors/help
Shared home/case list	Elder reference, latest result/time, escalation, overdue status, filters, urgent/newest first
Elder/case detail	Minimal profile, consent/eligibility, history, alert and escalation chain, activity interest
Case actions	Acknowledge, short note, follow-up complete, actor/time confirmation
Enrolment/sample record	Minimum fields, contacts, language, consent, activation validation
Test controls	Named testers only: demo call and sensor trigger, visibly labelled test-only
Analytics/health	Scenario results, timings, errors, unresolved cases, aggregate data

Admin	User access and approved activity sample

Navigation: sign in → shared case list → case detail → action. Urgent, worrying, and overdue items use text, icon, and colour. Mobile uses a single column, persistent back navigation, and large actions.

Accessibility and usability

- Traditional Chinese with natural Cantonese wording and English.
- Large readable text; **Recommendation:** at least 18 px body text and 44×44 CSS-pixel touch targets.
- **Recommendation:** WCAG 2.2 AA where practicable for prototype; formal review before pilot.
- Sufficient contrast, visible focus, semantic headings, labelled forms, logical tab order, descriptive errors, and screen-reader status announcements.
- Colour is never the only indicator.
- Prevent double submission and confirm consequential actions.

Usability targets: all scripted staff tasks complete without assistance in prototype testing; elder willingness and comprehension measured during pre-pilot research; usability issues affecting safety block pilot readiness.

7. NON-FUNCTIONAL REQUIREMENTS

Reliability and availability

The prototype has no production SLA. Core service should remain available through the demo window with a health check and documented restart. Calls, alerts, and webhooks use idempotency keys. Time-sensitive jobs persist across restarts or can be recovered from database state. Failures enter explicit failed/unresolved states and never fail silently.

Open Question: pilot service hours, on-call coverage, uptime target, maximum human response time, and formal service-level objectives require operator agreement.

Backup and disaster recovery

Prototype: GitHub code/config, database/sample export before demo, documented redeploy, and dated backup video. **Recommended prototype RPO:** 24 hours; **RTO:** four hours for sample data only. These are not safe commitments for live elder operations.

Future pilot: encrypted backups, tested restore, queue recovery, provider-outage plan, approved manual calling list, and agreed continuity owner.

Monitoring and alerts

Monitor API health, latency/errors, scheduler lag, overdue checks, call initiation/connection/webhook failures, sensor-to-call and concern-to-escalation times, stuck cases, chain exhaustion, login abuse, deletion failures, and provider usage/spend. Logs use request/case IDs and exclude phone numbers, audio, transcripts, and note text.

8. SUCCESS METRICS AND ANALYTICS

Objective	Metric	Prototype target	Future pilot use
Core reliability	Safe, worrying, no-answer consecutive passes	Each passes twice	Set operating threshold from evidence
Timely escalation	Worrying/second no-answer to first escalation start	100% within 60 seconds in acceptance tests	Track p50/p95 and every breach
Timely sensor response	Sensor event to elder call start	100% within 60 seconds	Track p50/p95 and breaches
Coverage	Check completed within about three days	Demonstrate schedule/overdue rule	Report exceptions and downtime
Avoid wasted checks	Safe verifications not requiring unnecessary manual visit	Demonstrate safe closure and notification	Compare staff effort with baseline
Do not hide risk	False-safe, missed, or delayed escalation	Zero in scripted set	Review every safety incident/near miss
Elder acceptance	Willing to try after in-person introduction	Not measured with team testers	≥70%; recommended ≥4/5 minimum sample
Staff value	Expect lower unnecessary workload without hidden risk	Not measured in prototype	Most interviewed staff
Community connection	Activity offered, interest, follow-up	Event captured	Establish baseline/target in pilot
Cost	Cost per completed check/escalation/enrolled elder	Record actual usage	Estimate price only after validation

Analytics events include profile activation, check due, call initiated/connected/ended, retry scheduled, sensor alert/call/closure, worrying detection, 999 guidance played, escalation steps, activity offered/interest/follow-up, staff acknowledgement/completion, opt-out, deletion, overdue check, and system/dependency error. Store category and timing only, not transcript or note content.

Review cadence: daily sprint review through Done/Next/Blocked; formal Day 6 readiness; immediate review of missed/delayed escalation or false-safe event; later weekly operational review and formal pilot evaluation.

9. IMPLEMENTATION PLAN

Six-day delivery plan

Assumption: Day 1 is 8 October and Day 6 is 13 October 2026. If the actual start differs, preserve the sequence.

Day	Product/business	Engineering	QA/evidence	Exit criterion
Day 1	Freeze journeys,	Telephony spike;	Consent test	Provider path or

	scripts, activity, personas, exclusions, approvers	scaffold UI/API/database/ adapters	numbers; evidence template	fallback; architecture/script s frozen
Day 2	Review Cantonese safety wording and bilingual labels	Profile/consent, call orchestrator, outcome states, safe activity, dashboard	Validation/state tests; first real connected call	Safe journey reaches dashboard
Day 3	Validate escalation wording and demo story	Worrying/unclear /immediate danger, one-minute chain, auth, staff actions	Worrying tests, timestamps, screenshots	Worrying journey and logins/actions work
Day 4	Review no-answer/sensor story	No-answer retry, sensor adapter, safe notification, bilingual/mobile, monitoring	**Internal checkpoint**, accessibility, regression, known issues; record backup demo	All flows integrated; if behind, change sensor to button/mock event
Day 5	Rehearse problem/value/li mits	Fix P0 only; stabilise deploy and recovery	Three core journeys pass twice; sensor timing; 3-user test; evidence bundle	Release candidate or named fallback
Day 6	Morning formal business + technical review	Freeze/deploy stable build; no unreviewed feature	Smoke test and backup video	Joint approval; live try, one retry, then disclosed recording

Roles and accountability

- **Ali:** software, architecture, integration, repository/deployment, technical approval.
- **Eden:** sensor interface, safe/no-answer testing, evidence.
- **Jovian:** telephony/contact-chain and worrying-response testing, evidence.
- **Abebi:** elder/staff research preparation, Cantonese/business content, stakeholder assumptions.
- **Jayson:** demo story, activity content, payer/value narrative, research support.
- **Recommendation:** formally name one of Abebi/Jayson as business lead and one of Eden/Jovian as independent release tester.

Collaboration standards

1. Every owner posts **Done / Next / Blocked** in WhatsApp daily.
2. Urgent safety, consent, external-service, timing, or test-number blocks are raised immediately.
3. Small paid services require recorded owner, purpose, spending limit, approval, and cancellation responsibility.

4. Final disagreements use a full-team vote. Safety/legal concerns cannot be voted away and block real use until qualified review.
5. Business and technical leads jointly approve the demo; either may withhold approval.
6. Stable-demo code requires review by another technical member and attached test evidence.
7. The formal readiness review is Day 6 morning; Day 4 is an internal risk checkpoint.

Recommended GitHub standard

- Protect `main` as demo-ready; use short branches such as `feat/call-retry` or `fix/duplicate-escalation`.
- Every issue lists requirement ID, owner, priority, dependencies, acceptance criteria, and test plan.
- PR title: `[FR-07] Implement no-answer retry`.
- PR body records change, privacy impact, evidence, known issues, and rollback.
- Author cannot be sole approver; one technical review required.
- Require passing type/lint/test checks and manual telephony evidence.
- No secrets, real elder data, unapproved numbers, recordings, or transcripts in Git.
- Tag the approved build and record deployed commit/test time.
- After Day 5 freeze, merge P0 fixes only, with focused regression evidence.

QA and evidence

Automate consent/eligibility, schedule/overdue, state machine, unclear→worrying, single retry timing, contact sequence, duplicate-event idempotency, permissions, conflict errors, deletion/no-audio, and privacy-safe analytics.

Manual scenarios cover safe, worrying, unclear, immediate danger, two no-answers, full contact chain, sensor-safe notification, unauthorised access, staff actions, desktop/mobile/bilingual/accessibility, no-audio inspection, three concurrent users, and demo fallback.

The evidence bundle contains checklist, timestamps, screenshots, provider event identifiers without private content, tester/reviewer, build commit, environment, actual vs expected result, defects, and known issues.

Defect severity and release

Severity	Meaning	Rule
P0 blocker	Missed/wrong/duplicate escalation; unsafe 999 claim; consent bypass; data/audio exposure; no real call; core scenario failure	Must fix and rerun regression; cannot approve live claim
P1 major	Required dashboard action broken, accessibility/bilingual blocker, missing timing evidence, failed sensor-safe notification	Fix before demo where possible; disclose only if core acceptance remains true

P2 minor	Cosmetic/content issue with safe workaround	Record and defer

Release: freeze → confirm review/tests/evidence → verify sample data, test numbers, recording disabled and spend → smoke test and backup video → business/technical joint approval → tag build and known issues → live attempt, one retry, then disclosed recording.

Rollout

1. **Prototype:** fake data, team numbers, simulated adapters, ≤3 users.
2. **Readiness:** research ≥5 elders and ≥2 staff; complete legal/privacy/telecommunications/safeguarding review; assign owner, hours, and incident process.
3. **Controlled pilot:** 20–30 eligible elders through a consenting organisation; phased enrolment and daily early review.
4. **Evaluation:** safety, coverage, timings, workload, acceptance, activity connection, cost, and provider performance.
5. **Procurement/scale:** estimate price and select charging model only after evidence; then add production integrations, resilience, tenancy, or languages.

10. RISK ASSESSMENT AND MITIGATION

Risk	Probability	Impact	Prevention	Contingency	Early warning	Owner/response
Cantonese misunderstood	High	High	Day 1 spike, fixed phrases, unclear→worrying, Cantonese review	Simpler script/menu; block pilot until validation	Repeats or tester disagreement	Ali stops flow; testers capture; business revises script
Provider lacks HK/landline/fixed caller capability	Medium–High	High	Validate Day 1	Approved alternate if feasible; disclosed backup	Verification delay/caller mismatch	Ali decides provider path
Escalation delayed or duplicated	Medium	Critical	Durable jobs. idempotency state tests	Manual contact in tests; block live claim	Queue lag, duplicate events	Ali fixes; Eden/Jovian rerun
Unsafe medical or 999 claim	Low–Medium	Critical	Constrained scripts and prohibited-content tests	Block release; replace wording	Content-test failure	Business + technical leads block
Scam suspicion	High	High	In-person introduction, fixed number, disclosure,	Known-centre human follow-up; opt-out	Hang-ups, identity questions	Abebi/Jayson refine onboarding

			no money/ID			
Consent/privacy/telecom breach	Medium	Critical	Three-party consent, fake data, no audio, access control, required reviews	Stop calls/access; contain; follow approved incident process	Unknown data/recording/access	Ali disables; organisation manages incident
Data routed outside acceptable region	Medium	High	Provider/sub processor review and minimisation	Replace/restrict provider; no real data	Unclear storage terms	Privacy/legal owner decides
Chain ends without staffed operator	Medium	Critical	Confirm roster and hours	Keep unresolved; approved manual escalation	Repeated chain exhaustion	Operating organisation responds
Sensor work consumes sprint	High	Medium	Adapter and simulation first	Day 4 switch to button/mock event	Not integrated at checkpoint	Ali cuts scope; Eden/Jovian verify
Dashboard/call fails in demo	Medium	Medium	Freeze, smoke test, health check, backup video	Retry once then disclosed recording	Pre-demo errors	Ali operates fallback
Real data enters prototype	Low–Medium	High	Seed data and number allow-list	Stop/remove/rotate/review	Unknown name/number in logs	Any member blocks; Ali contains
False safe hides risk	Medium	Critical	Conservative classification and human incident review	Escalate uncertainty; pause rules/pilot	Later contradiction	Care/safeguarding lead investigates
Workload increases	Medium	Medium–High	Baseline measurement and shared queue	Reduce cohort/adjust workflow	Backlog/false alerts	Product owners review with staff
Six-day delivery slips	Medium–High	High	P0-first, daily blocks, Day 4 gate	Mock sensor; fixed activity; no live partner integration	Missing flow by Day 4	Team votes cut; Ali protects core

Pricing assumed without evidence	High	Medium	Track actual use/cost and buyer interviews	Keep price open	Requests for unsupported price	Abebi/Jayson label and validate
Excluded users enrolled	Medium	High	Eligibility block and training	Remove and refer to suitable service	Cohort flag	Referring organisation owns suitability

11. APPENDIX

Assumptions

1. The six-day sprint runs 8–13 October 2026; dates change if actual start differs.
2. Prototype calls use only consenting team numbers.
3. A managed service may support real Hong Kong calls; Day 1 must prove it.
4. Existing manual workflows are fragmented/time-consuming; staff research must confirm.
5. Regular respectful contact may contribute to connection and earlier awareness; no mortality or isolation reduction is claimed.
6. One approved sample activity is enough for the prototype.
7. Team capacity is **Estimate:** 60–120 person-hours.
8. One shared authorised dashboard can serve the three staff groups in the prototype.
9. The operator number is staffed; this must be confirmed before real use.
10. Supplied population, pilot-coverage, and centre figures require citation and definition checking in external publication.

Open questions

Safety and operations: exact approved Cantonese wording; pilot hours and roster; incident stop criteria; definition of "checked"; detection of human acknowledgement.

Users/research: whether four of five elders is the minimum; staff decision rule with two interviewees; eligibility and referral criteria; accommodations for mild hearing/cognitive needs.

Technology: provider; immediate-danger call interruption; partner interfaces; Hong Kong hosting across subprocessors; final SLO/rate/RPO/RTO/load targets.

Privacy/legal: form of consent evidence; legal basis/notices/access/correction/cross-border/breach terms; automated-call and caller-ID rules; allowable post-deletion audit; data controller and processor roles.

Business: exact government buyer and budget owner; future charging unit; actual cost; validated future price range and procurement path; owner of activity suitability and availability.

Plain-language glossary

Term	Meaning
AI disclosure	Clearly telling the person that the caller is

	automated AI
Check-in	Scheduled or alert-triggered call asking whether the elder is safe
Safe response	Clear indication of safety; creates short note and activity offer
Worrying response	Help request, unsafe statement, or unclear/confused answer
Immediate danger	Urgent statement; advise 999 and alert a human immediately
No answer	No connected human response; after one retry it escalates
Escalation	Calling first contact, second contact, then operator for human action
Human-in-the-loop	A person, not AI, decides emergency action
Sensor verification	AI call after an existing sensor alert
Short outcome note	Brief status/follow-up summary, not transcript, diagnosis, or audio
Hidden elder	Older person with limited formal/community contact; exact definition needs validation
Prototype	Six-day demo using fake data and team numbers
Pilot	Future controlled trial planned for 20–30 eligible elders
Adapter	Software boundary giving simulated and partner systems one interface
Idempotency	Repeated event processing without duplicate calls or escalations
RPO / RTO	Acceptable data loss / target recovery time after disruption
P0 / P1 / P2	Blocker / major / minor defect levels
MoSCoW	Must, Should, Could, Won’t priority method
Data minimisation	Collecting and keeping only what the service genuinely needs

Prototype acceptance summary

The prototype is acceptable for demonstration only when real calls reach consenting team phones; the three core journeys each pass twice; one-minute initiation targets and 5–10-minute retry are evidenced; the full contact chain works; sensor-safe closure informs a human; dashboard fields/actions/login/bilingual/responsive/accessibility requirements pass; no real elder data or audio exists; evidence and known issues are recorded; stable code is peer reviewed; and business and technical leads jointly approve on Day 6 morning.

Passing the prototype does not authorise real elder calls. Real use additionally requires three-party agreement, target-user research, legal/privacy/telecommunications/safeguarding review, provider validation, operating ownership, and approved pilot procedures.

Want me to adjust any section, simplify any part, or list the assumptions to double-check?